

Tracking Community Resilience (TraCR) Database

Technical Documentation

Version 1.0

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Tiffany Cousins, National Institute of Standards and Technology
Donghwan Gu, National Institute of Standards and Technology & University of Maryland,
College Park
Maria Dillard, National Institute of Standards and Technology

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Introduction

Community resilience is the ability to prepare for disruptions, adapt to changing conditions, and withstand and recover within a specified time frame from disruptions (NIST, 2015). It is a complex, multi-dimensional problem that relies on engineering, social sciences, earth sciences, and other disciplines to improve the way communities prepare for, resist, respond to, and recover from disruptive events, whether those events are due to natural or human-caused hazards (NIST, 2020). With a growing field of research focused on community-level resilience, many entities have developed and examined approaches to measure resilience both quantitatively and qualitatively (Loerzel & Dillard, 2021).

The National Institute of Standards and Technology (NIST) Community Resilience Program is focused on developing science-based tools and measures to support community resilience planning and implementation. Specifically, the Program includes projects focused on quantifying community resilience, supporting identification and evaluation of alternative resilience solutions, and engaging community resilience stakeholders for input and feedback on research and products (e.g., guidance, tools, and measures). The NIST Community Resilience Program works closely with NIST's Disaster and Failures Studies (DFS) Program to develop data and metrology that support validation of community resilience models and tools.

The Community Resilience Assessment Methodology project set out to develop tools and measures to quantitatively measure resilience at the community scale. The goal of this project is to develop a resilience assessment methodology that can be applied to communities to assess baseline resilience and changes in resilience over time. The assessment methodology employs a complex systems perspective to make linkages between social and physical systems and aims to address the dynamic aspects of resilience over time.

There are two foundational products developed by the NIST Community Resilience Assessment Methodology team to support the measurement science needed to create a validated methodology to assess resilience at the community scale. The first product is the Inventory of Community Resilience Indicators & Assessment Frameworks (Loerzel & Dillard, 2021). The goal of the Inventory was to gather, analyze, and document existing frameworks for the purposes of developing a community resilience measurement methodology. The Inventory takes the form of a database which contains 56 existing quantitative resilience frameworks, 3,298 indicators, and 7,165 measures that have been evaluated and catalogued according to a standardized methodology (Loerzel & Dillard, 2021). The second product is the Tracking Community Resilience (TraCR) database, a data compilation tool for multi-system, county-level, longitudinal data across the United States.

Tracking Community Resilience (TraCR)

Communities interested in improving their resilience to hazards and other disruptions require decision-support and assessment tools to gather baseline data, evaluate the impact and effectiveness of programs and investments, and monitor changes over time. However, there is a lack of consensus with respect to the theoretical approach taken, indicators used, data needs, and scale of application among resilience assessment methodologies and tools. The TraCR database was developed initially for NIST researchers as a sandbox for developing and testing community resilience measures. Through this research, it was found that there are very few physical system measures (Gerst et al., 2024). This finding served as a basis for building out the TraCR database to include physical systems indicators. As of September 2025, the version 1.0.0 database contains 101 selected measures from the main TraCR database, primarily consisting of secondary data reorganized and treated by NIST.

The TraCR database provides measures and data values for counties across all 50 states and selected U.S. territories (Puerto Rico and the U.S. Virgin Islands). Counties were chosen as the unit of analysis because they allow for better statistical comparisons, especially for indicators conceptually rooted in larger geographic areas, such as regional economic vulnerability and healthcare infrastructure (Gu et al., 2025). The version 1.0 database is an unbalanced panel data containing 3,232 counties across the period from 2000 to 2021. The longitudinal data availability is presented in [Appendix A - Data Coverage].

TraCR can be used:

- By local communities, to summarize key measures and trends across major sectors, benchmarked against adjacent communities.
- In grant applications, to provide data and evidence for assessing the projected role and impact of the grant.
- And, by researchers, to establish baseline measures, including database attributes like multi-source longitudinal data accessibility.

Domains

In version 1.0.0 of the TraCR, there are 101 measures across 6 domains. The use of domains is intended for the organization of measures within the database, not as a structure for measuring community resilience (Table 1). These are Infrastructure, Built Environment, Social Well-Being, Natural Environment, Economic Vitality, and Recovery of Community Function. Sub-domains are assigned to each domain, and measures are assigned to each sub-domain. (Figure 1).

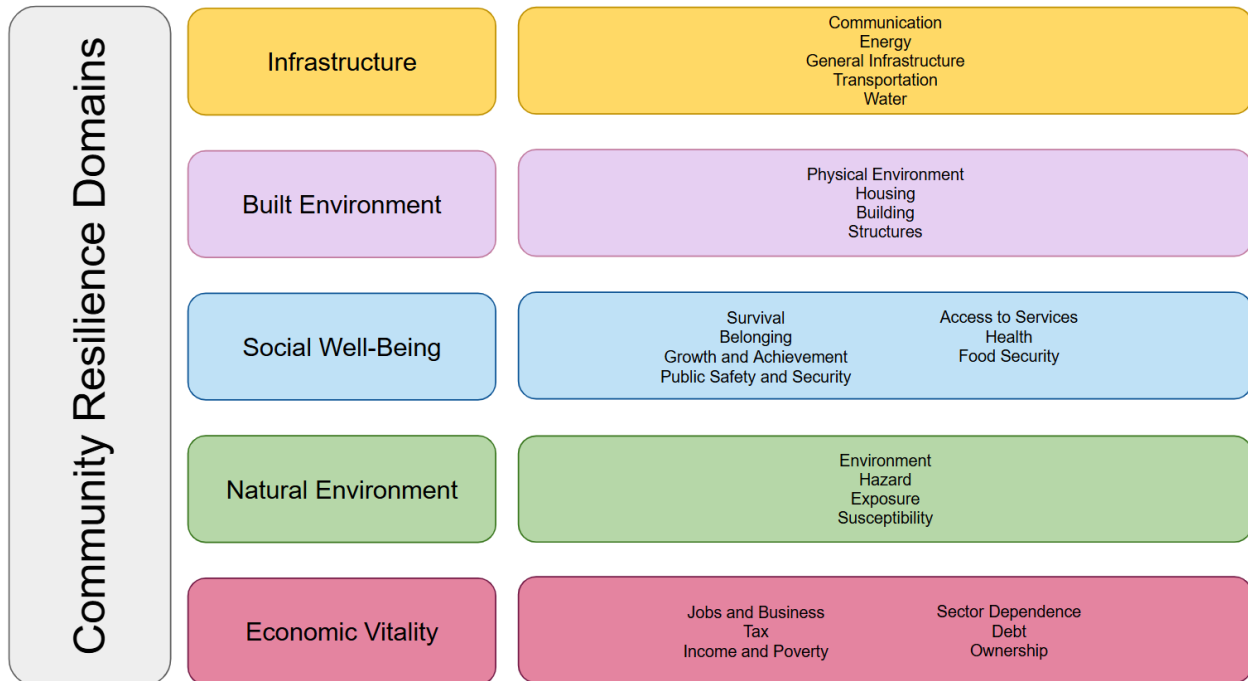


Figure 1: TraCR community resilience domains

Community Resilience Domain Codes

For ease of use, measures are given a variable code based on their home domain type and indicator. Measures can be used outside of their home system type and indicator. For example, the metric “estimate of unemployed persons” can be used to measure *Income and Poverty* under Economic Vitality and *Survival* under Social Well-Being.

100000	Infrastructure (INFRA)
110000	Communication
120000	Energy
130000	General Infrastructure
140000	Transportation
150000	Water
200000	Built Environment (BENV)
210000	Physical Environment
220000	Housing
230000	Buildings
240000	Structures
300000	Social Well-Being (SWB)

310000	Survival
320000	Belonging
330000	Growth and Achievement
340000	Public Safety and Security
350000	Access to Services
360000	Health
370000	Food Security

400000	Natural Environment (NENV)
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410000	Environment
420000	Hazard
430000	Exposure
440000	Susceptibility

500000	Economic Vitality (ECNV)
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510000	Jobs and Businesses
520000	Tax
530000	Income and Poverty
540000	Sector Dependence
550000	Debt
560000	Ownership

Measures

As of September 2025, TraCR version 1.0.0 (TraCR v1.0.0) has been released. This version is a flat file database (i.e., a tabular file without an internal hierarchy between records) consisting of 101 measures. Within the database, no data or null are represented as blank cells. Please see the metadata documentation file included in the downloaded data package for detailed information on data sources and calculation methods.

Measure ID	Description	System Type	Indicator
INFRA110007	% of households with broadband internet service	Infrastructure	Communication
INFRA120001	SAIDI (System Average Interruption Duration Index) with MED (Major Event Days)	Infrastructure	Energy
INFRA120002	SAIDI (System Average Interruption Duration Index) without MED (Major Event Days)	Infrastructure	Energy
INFRA120003	SAIFI (System Average Interruption Frequency Index) with MED (Major Event Days)	Infrastructure	Energy
INFRA120004	SAIFI (System Average Interruption Frequency Index) without MED (Major Event Days)	Infrastructure	Energy
INFRA120005	% of housing units heated by electricity	Infrastructure	Energy
INFRA120006	% of housing units heated by fuel piped to site	Infrastructure	Energy
INFRA120007	Year-over-year increase in homes heated by solar	Infrastructure	Energy
INFRA120008	% of housing units heated by fuel trucked to site	Infrastructure	Energy
INFRA130002	NOAA Critical Facilities: Hospital Medical	Infrastructure	Infrastructure
INFRA140003	Average AADT of bridges that are at least 50 years old and in poor condition	Infrastructure	Transportation
INFRA140004	Average detour of bridges that are at least 50 years old and in poor condition	Infrastructure	Transportation
INFRA140007	Mean travel time to work (mins)	Infrastructure	Transportation
INFRA150001	% served by ground water public water systems	Infrastructure	Water
INFRA150002	% of population served by ground water Public Water Systems that are out of compliance	Infrastructure	Water
INFRA150003	% of population served by ground water Priority Public Water Systems Returned to Compliance	Infrastructure	Water
INFRA150004	% of population served by ground water Public Water Systems Identified as Priority Systems	Infrastructure	Water
INFRA150005	% of population served by ground water Public Water Systems returned to compliance	Infrastructure	Water
INFRA150006	% of population affected by ground water Public Water Systems with at least 1 violation	Infrastructure	Water
INFRA150007	% served by surface water public water systems	Infrastructure	Water
INFRA150008	% of population served by surface water Public Water Systems that are out of compliance	Infrastructure	Water
INFRA150009	% of population served by surface water Priority Public Water Systems Returned to Compliance	Infrastructure	Water

INFRA150010	% of population served by surface water Public Water Systems Identified as Priority Systems	Infrastructure	Water
INFRA150011	% of population served by surface water Public Water Systems returned to compliance	Infrastructure	Water
INFRA150012	% of population affected by surface water Public Water Systems with at least 1 violation	Infrastructure	Water
INFRA150013	% of major facilities	Infrastructure	Water
INFRA150014	% of major facilities in significant noncompliance (SNC)	Infrastructure	Water
INFRA150015	% of major facilities with violations	Infrastructure	Water
INFRA150016	% of non-major facilities	Infrastructure	Water
INFRA150017	% of non-major facilities in Category I Noncompliance (SNC equivalent)	Infrastructure	Water
INFRA150018	% of non-major facilities with violations	Infrastructure	Water
BLTENV220001	% of housing stock built to modern building codes (IRC) (using NSI data)	Built Environment	Housing
BLTENV220002	% of housing units that are less than 15 years old (using NSI data)	Built Environment	Housing
BLTENV220003	% of housing units that are more than 30 years old (using NSI data)	Built Environment	Housing
BLTENV220004	Average year built of housing stock (using NSI data)	Built Environment	Housing
BLTENV220005	% of housing units that are less than 15 years old (using census data)	Built Environment	Housing
BLTENV220006	% of housing units that are more than 30 years old (using census data)	Built Environment	Housing
BLTENV220007	Average year built of housing stock (using census data)	Built Environment	Housing
BLTENV220008	% of housing stock built to modern building codes (IRC) (using census data)	Built Environment	Housing
BLTENV220009	% of housing units that are mobile homes	Built Environment	Housing
BLTENV220010	% of housing units that are owner-occupied	Built Environment	Housing
BLTENV220011	% of Units available for rent (excluding seasonal, rec, occ, migrant & other vacant)	Built Environment	Housing
BLTENV220017	US Census Households: without Telephone Service (1-Year Estimate)	Built Environment	Housing
BLTENV220018	US Census Households: without Telephone Service (5-Year Estimate)	Built Environment	Housing
BLTENV220022	US Census Housing Units: Mobile Home (1-Year Estimate)	Built Environment	Housing
BLTENV220023	US Census Housing Units: Mobile Home (5-Year Estimate)	Built Environment	Housing
BLTENV220029	US Census Housing Units: Vacant (1-Year Estimate)	Built Environment	Housing
BLTENV220030	US Census Housing Units: Vacant (5-Year Estimate)	Built Environment	Housing
SOCWB310003	US Census Households: with > 60-Year-Old (1-Year Estimate)	Social Well-Being	Survival
SOCWB310004	US Census Households: with > 60-Year-Old (5-Year Estimate)	Social Well-Being	Survival
SOCWB310005	US Census Households: No Vehicle Available (1-Year Estimate)	Social Well-Being	Survival

SOCWB310006	US Census Households: No Vehicle Available (5-Year Estimate)	Social Well-Being	Survival
SOCWB310007	US Census Housing Units: Median monthly costs of housing for all owner-occupied units (with mortgage) (1-Year Estimate)	Social Well-Being	Survival
SOCWB310008	US Census Housing Units: Median monthly costs of housing for all owner-occupied units (with mortgage) (5-Year Estimate)	Social Well-Being	Survival
SOCWB310009	US Census Housing Units: Median monthly costs of housing for all owner-occupied units (without mortgage) (1-Year Estimate)	Social Well-Being	Survival
SOCWB310010	US Census Housing Units: Median monthly costs of housing for all owner-occupied units (without mortgage) (5-Year Estimate)	Social Well-Being	Survival
SOCWB310011	US Census Housing Units: Median Rent (1-Year Estimate)	Social Well-Being	Survival
SOCWB310012	US Census Housing Units: Median Rent (5-Year Estimate)	Social Well-Being	Survival
SOCWB310015	US Census Population: Estimate	Social Well-Being	Survival
SOCWB310017	US Census Population: 65 years and older	Social Well-Being	Survival
SOCWB310018	US Census Population: Under 18 Years (1-Year Estimate)	Social Well-Being	Survival
SOCWB310019	US Census Population: Under 18 Years (5-Year Estimate)	Social Well-Being	Survival
SOCWB320013	US Census Ethnicity: Hispanic or Latino	Social Well-Being	Belonging
SOCWB320016	US Census Race: Asian	Social Well-Being	Belonging
SOCWB320017	US Census Race: American Indian	Social Well-Being	Belonging
SOCWB320018	US Census Race: Black	Social Well-Being	Belonging
SOCWB320019	US Census Race: White	Social Well-Being	Belonging
SOCWB330004	US Census Population: 25 years and older with a bachelor's degree or higher (1-Year Estimate)	Social Well-Being	Growth and Achievement
SOCWB330005	US Census Population: 25 years and older with a bachelor's degree or higher (5-Year Estimate)	Social Well-Being	Growth and Achievement
SOCWB330006	US Census Population: 25 years and older with a GED or higher (1-Year Estimate)	Social Well-Being	Growth and Achievement
SOCWB330007	US Census Population: 25 years and older with a GED or higher (5-Year Estimate)	Social Well-Being	Growth and Achievement
SOCWB330008	US Census Population: No High School Diploma/GED (1-Year Estimate)	Social Well-Being	Growth and Achievement
SOCWB330009	US Census Population: No High School Diploma/GED (5-Year Estimate)	Social Well-Being	Growth and Achievement
SOCWB330010	US Census Population: Individuals in Poverty (1-Year Estimate)	Social Well-Being	Income and Poverty
SOCWB330011	US Census Population: Individuals in Poverty (5-Year Estimate)	Social Well-Being	Income and Poverty
SOCWB330012	US Census Population: Under 18 in Poverty (1-Year Estimate)	Social Well-Being	Income and Poverty
SOCWB330013	US Census Population: Under 18 in Poverty (5-Year Estimate)	Social Well-Being	Income and Poverty

SOCWB330014	US Census School Enrollment: Grades K-12 (1-Year Estimate)	Social Well-Being	Growth and Achievement
SOCWB330015	US Census School Enrollment: Grades K-12 (5-Year Estimate)	Social Well-Being	Growth and Achievement
SOCWB330016	US Census School Enrollment: Public School Grades K-12 (1-Year Estimate)	Social Well-Being	Growth and Achievement
SOCWB330017	US Census School Enrollment: Public School Grades K-12 (5-Year Estimate)	Social Well-Being	Growth and Achievement
SOCWB350001	Number of nursing home and assisted living facility beds per 10,000 people over 85 years old	Social Well-Being	Access to Services
SOCWB350002	% of population within 60 minutes of Level 1 Trauma center by air or ground transportation	Social Well-Being	Access to Services
SOCWB350003	% of population within 60 minutes of Level 3 Trauma center by air or ground transportation	Social Well-Being	Access to Services
SOCWB350006	US Census Population: No Health Insurance (1-Year Estimate)	Social Well-Being	Access to Services
SOCWB350007	US Census Population: No Health Insurance (5-Year Estimate)	Social Well-Being	Access to Services
NATENV410002	NLCD: Percent of land cover that is Agricultural	Natural Environment	Environment
NATENV410003	NLCD: Percent of land cover that is Developed	Natural Environment	Environment
NATENV410004	NLCD: Percent of land cover that is Vegetated	Natural Environment	Environment
NATENV410005	NLCD: Percent of land cover that is Wetland	Natural Environment	Environment
ECNVIT510001	BLS Civilian Labor Force Unemployment Rate	Economic Vitality	Jobs and Businesses
ECNVIT510002	US Census Population: Unemployed (1-Year Estimate)	Economic Vitality	Jobs and Businesses
ECNVIT510003	US Census Population: Unemployed (5-Year Estimate)	Economic Vitality	Jobs and Businesses
ECNVIT530001	US Census Households: Income over \$200,000 (1-Year Estimate)	Economic Vitality	Income and Poverty
ECNVIT530002	US Census Households: Income over \$200,000 (5-Year Estimate)	Economic Vitality	Income and Poverty
ECNVIT530003	US Census Households: Poverty (1-Year Estimate)	Economic Vitality	Income and Poverty
ECNVIT530004	US Census Households: Poverty (5-Year Estimate)	Economic Vitality	Income and Poverty
ECNVIT530005	US Census Housing Units: Median Household Income (1-Year Estimate)	Economic Vitality	Income and Poverty
ECNVIT530006	US Census Housing Units: Median Household Income (5-Year Estimate)	Economic Vitality	Income and Poverty
ECNVIT530007	US Census Per Capita Income (1-Year Estimate)	Economic Vitality	Income and Poverty
ECNVIT530008	US Census Per Capita Income (5-Year Estimate)	Economic Vitality	Income and Poverty

Planned Updates

Updates to TraCR will be phased to allow expansion of the measures and time periods for which data are available. The phased approach will allow for user feedback and continued

research on the most useful community resilience measures. Updates are categorized as major, minor, and patch. Major updates consist of new features or products (i.e., publishing a web mapping application for TraCR). NIST expects to make major updates on a two-year cycle, as resources allow. Minor updates consist of releasing additional measures. NIST expects to make minor updates annually, as resources allow. Patch updates consist of fixing data value errors. Patch updates will be released as needed.

Data Sources and Descriptions

TraCR v1.0.0 has 101 measures compiled from eight sources. This section describes the data sources and variables that form these measures. The metadata documentation includes source information for each measure. Below, data sources are listed with a data acknowledgement statement and the period of data used. Unless stated otherwise, all data acknowledgements are reproduced verbatim from their original sources.

1. Bureau of Labor Statistics (BLS)

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- Time period of data used: 2000-2016

2. United States Census Bureau (CEN)

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- Time period of data used: ACS 1-Year and 5-Year Estimates, ranging from 2000-2016

3. United States Energy Information Administration (EIA)

U.S. Energy Information Administration (June 2025).

- Time period of data used: 2010-2021

4. Environmental Protection Agency (EPA)

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- Time period of data used: 2020

5. Federal Communications Commission (FCC)

“This product may use the FCC Data API but is not endorsed or certified by the FCC.” (FCC API Terms of Service, 2025)

- Time period of data used: 2014-2021

6. Homeland Infrastructure Foundation-Level Data (HIFLD)

The datasets in TraCR are for Public Use.

- Time period of data used: 2023

7. National Oceanic and Atmospheric Administration (NOAA)

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- Time period of data used: 2015

8. National Structures Inventory (NSI)

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- Time period of data used: 2023

Supplemental Files

The metadata Excel file provides a structured overview of the dataset version 1.0.0. It contains the measures with source information, data schema, and data descriptions.

Appendix A Data Coverage

The data availability matrix Excel file visually represents the coverage of the dataset version 1.0.0. This matrix presents the number of observations for each measure across the period from 2000 to 2021. The cells are color-coded to indicate an absence of data (highlighted in grey), and cells with a substantial number of observations (highlighted in blue)—over 3,000 observations among the total of 3,232 observations.

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